

Client Report

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Client(s): Cornell CALS Extension / E&J Gallo Winery / National Grape

Context and Problem Statement:

During short, high-volume harvest periods, facilities must remove spotted lantern flies (SLFs) quickly, yet current methods continue to miss insects, risking contamination and regulatory noncompliance, which can lead to financial and operational strain. There are three opportunities to remove SLF: before, during, and after harvest processing. Our approach is implemented during the harvesting stage, when there is minimal grape juice on the conveyor. Operating at this point minimizes product loss, improves removal efficiency, and maintains compatibility with high-throughput processing conditions. The conveyor is also a single point where all harvested grapes pass through. By looking for solutions to apply at this stage, we reduce the number of devices needed to effectively remove SLF.

Final Prototype and Application:

Our concept is a filtering brush mechanism designed to remove SLF from grapes during the harvesting process. Grapes will travel along the harvesting machine's conveyor belt and pass through a rotating brush placed above. The brush will lightly contact the product stream, applying enough force to knock off the SLF from the grapes. Our prototype is designed to integrate directly with the conveyor rails via mounting brackets specifically fitted to the conveyor dimensions. It can be installed as an inline module within the harvesting process, operating continuously as grapes are transported through the processing line.



Figure 1. Conveyor belt processing grapes during harvesting. Image found from harvester videos shared by Terry Bates.

Success Criteria:

Our design was a rotary brush tool that can be attached to current harvester conveyor belts to remove SLF from grapes just harvested from vines. The criteria we sought out was to ensure:

1. The brushes should not damage, crush, or visibly harm more than 5% of grapes. Our clients have emphasized product loss as a major problem in current SLF removal systems.
2. The brushes should be tuned to remove at least 90% of SLF from grapes in a single pass. High removal rates are a high priority to prevent product rejection.
3. The system should be able to withstand realistic operation without shifting more than 5mm. Greater stability is a medium priority.
4. The system should be able to process grapes on a conveyor belt at a speed of 1 m/s. Higher speeds are a medium priority.

Testing and Results:

Our initial prototype consisted of food grade brushes rotating on a wooden dowel that is driven by a standard hand held drill. It is mounted above a conveyor system to simulate harvesting conditions. The main mechanical risks involve brush motion and structural stability. The following tests allowed us to identify three factors which strongly impacted performance: brush length, rotation speed, and rotation direction.



Figure 2. Initial Prototype of Mount and Brush Mechanism. The wooden dowel is driven by a handheld drill.

During testing we made a mock conveyor belt and modeled grapes out of clay, each of the approximate weight of a grape (5-7g). We modeled SLF with loops of masking tape, approximating their weight and clinging tendency. Using the cordless drill to drive the brushes, we measured how many SLF were knocked off and how much our mounting moved. By changing the direction of the drill, we also tested whether a clockwise or counterclockwise brush rotation was more effective for SLF removal.

Drill Spec Test - 3" Brush		
Direction \ Speed	500 RPM	2000 RPM
Clockwise	3/8	6/8
Counterclockwise	0/8	0/8

Table 1. Speed and direction of brush rotation greatly impacts effectiveness. 2000 RPM was the most successful speed.

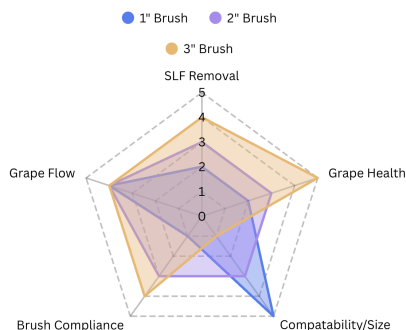


Table 2. Qualitative results of Brush Length Test. The 3" brush is the most well-rounded choice.

Our testing highlighted several areas in need of improvement. For drill speeds >500 RPM, our prototype started shifting, indicating a need for a sturdier, more robust mounting design. As we deduced that 2000 RPM in the clockwise direction, as that least impacted the flow of the grapes, removed the most SLF, we needed sturdier mounts on future prototypes. We measured the grape flow and grape health more qualitatively, relative to each other. Unfortunately we could not simulate enough tests to come up with concrete numbers, however the testing was still fruitful. The 3" brush was identified as the most effective choice, as shown in Table 2, which should also be integrated in future configurations.

Final Prototype:

The initial prototype established baseline estimates for dimensions, operating speed, and overall feasibility. The final prototype builds on this by closely replicating a real harvester conveyor system with an integrated rotary brush mechanism. A mock conveyor section was constructed with dimensions similar

to an actual harvesting belt, using food-grade materials for the belt and brushes, along with 3D-printed components. The mounts are machined from aluminum, enabling precise placement within a 0.010-inch tolerance and ensuring structural stability. The system is driven by a 3000 RPM motor paired with a speed controller, allowing fine adjustment of operating speed to meet specific testing requirements.

Reevaluation of mounting stability showed that for 2000 RPM operating speeds, the mounting did not shift. Additionally, the final prototype incorporates a shaft with an internal helical channel for mounting the brushes. This design increases stability by maximizing the brush mounting surface area. In contrast, the first prototype used four externally attached rod supports (shown in Figure 2), whereas the final design provides a continuous mounting interface.

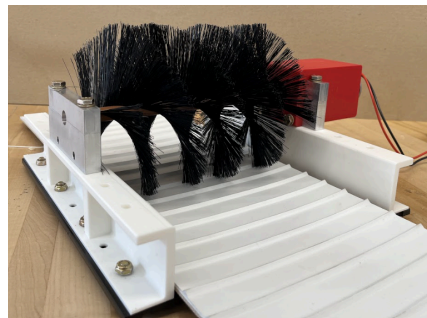


Figure 3. Final Prototype Assembly. The mounts are aluminum while the rod and motor casing are 3D printed.

Conclusion and Recommendation:

We researched the already existing harvesters and found the OXBO 6030 was already equipped with PREMIUMDESTEM and PREMIUMSORT technology that included a majority of what our product was trying to achieve. Therefore, as our results prove that the brush mechanisms are indeed brushing off SLF with increased success and efficiency, further improvements can be made to the already established technology.

- A STANDARD FAN**
Removes MOG (material other than grape) from fruit stream
- B DESTEMMER**
Separates grapes from the rachis.
- C RACHIS DISPOSAL**
Rachis drops to the vineyard floor.
- D PETIOLE BELT**
Using a patented ratcheted action, singulated grapes pass through the petiole belt while petioles, jacks, and small debris are taken to the trash auger. For the cleanest, straight-to-tank quality fruit.
- E RETURN BELT**
Returns clean, sorted grapes to the cross conveyor.
- F TRASH AUGER**
Drops petioles and small debris to the vineyard floor.
- G CROSS CONVEYOR**
Clean grapes move to the over the row conveyor or bin.



Future iterations should explore incorporating multiple brush stages with varying stiffness or rotational speeds to improve removal efficiency while minimizing damage to the grapes. Introducing a staged approach would allow for more controlled, gradual removal of SLF across the process. A practical implementation should include an integrated cleaning mechanism to prevent debris buildup and brush clogging, ensuring consistent performance and reducing maintenance requirements over time. Software-based control can also improve performance significantly; for example, a closed-loop feedback system, using vision-based sensors, could enable real-time adjustment brush speeds/torque based on SLF presence.

Figure 4. Components of existing debris removal technology.
(Image found from OXBO.com about 6030)

References

Figure 1: <https://cornell.app.box.com/v/harvestVideos/file/2170562711012>

Figure 4: https://www.oxbo.com/app/uploads/2022/06/6030_25_EN_USLTR_RGB.pdf

Bill of Materials

Total Budget:	\$350.00	Current Spending:	\$334.94	Remaining Budget:	\$15.07
McMaster/Amazon/Machined	Product	Quantity	Unit Cost	Total Cost	Link
Amazon	2PCS LM25UU Linear Ball Bearings, 25mm Bore Dia, 40mm OD, 59mm Length with Double Side Rubber Seal Linear Rail for CNC,3D Printer	1	\$15.99	\$15.99	weawert-LM25UU-Linear-Bearin
McMaster	Oak Rod, 36" Long, 1" Diameter	1	\$10.79	\$10.79	96825K82
McMaster	Steel Ball Bearing Open, Trade Number R16, for 1" Shaft Diameter	2	\$11.73	\$23.46	60355K509
McMaster	(Length = 5 ft) Food Industry Easy-Cut Strip Brush 1/8" Wide x 3/32" High Backing, 1" Overall Height	1	\$17.50	\$17.50	1469N21
McMaster	(Length = 5 ft) Food Industry Easy-Cut Strip Brush 1/8" Wide x 3/32" High Backing, 2" Overall Height	1	\$18.75	\$18.75	1469N22
McMaster	(Length = 5 ft) Food Industry Easy-Cut Strip Brush 1/8" Wide x 3/32" High Backing, 3" Overall High	2	\$21.25	\$42.50	1469N23
McMaster	1/4" acrylic sheet	2	\$7.00	\$14.00	
McMaster	0.5" aluminum rod	0.01	\$5.00	\$0.05	
McMaster	Zinc Yellow-Chromate Plated Hex Head Screw Grade 8 Steel, 1/4"-28 Thread, 1-1/4" Long, Partially Threaded	0.04	\$13.65	\$0.55	91257A565
McMaster	Zinc Yellow-Chromate Plated Hex Head Screw Grade 8 Steel, 1/4"-28 Thread, 1/2" Long, Partially Threaded	0.08	\$21.21	\$1.70	92620A562
McMaster	Zinc Yellow-Chromate Plated Hex Head Screw Grade 8 Steel, 1/4"-28 Thread, 5/8" Long, Partially Threaded	0.08	\$14.89	\$1.19	92620A563
McMaster	"Steel Ball Bearing Open, Trade Number R8, for 1/2" Shaft Diameter"	4	\$11.73	\$46.92	60355K509
McMaster	shaft coupler	1	\$14.15	\$14.15	6412K12
Amazon	12V 30W Permanent Magnet DC Motor, 3000RPM High Speed CW/CCW Electric Gear Motor for DIY Generator and 12V-40V 10A PWM DC Motor Speed Controller Fan Controller with Knob-High Efficiency	1	\$24.99	\$24.99	vS_YsU3gO4C3gp3wtq_-68Now
McMaster	White Cleated Belting for Slider-Bed Conveyors, Buna-N, 8" wide, 2' long	1	\$46.26	\$46.26	5987K1
3D Printed	Walls	2	\$0.05	\$9.18	
3D Printed	Motor Mount	1	\$0.05	\$3.84	
3D Printed	Rod with Internal Channels	1	\$0.05	\$1.44	
Machined	Lower Wall Mount	2	\$0.28	\$0.56	
Machined	Upper Wall Mount	2	\$0.56	\$1.12	
Machined	Hours of Machining Time	8	\$5.00	\$40.00	